Capacitors: a short learning note

Page 1 - Charge and voltage

A capacitor stores charge on two separated conductors.

Capacitance is the ratio of stored charge to voltage: C = Q / V.

Energy stored in a capacitor

Page 2 - Energy

Energy is stored in the electric field.

The stored energy is U = 1/2 C V^2.

Increasing the voltage requires more work for each added charge.

Practice and limits

Page 3 - Practice

At fixed capacitance, doubling voltage quadruples stored energy.

This note does not discuss quantum computing or battery chemistry.